

Drug-synergy prediction under honest pair-held-out evaluation, and AND-gated biclonal CAR-T target discovery from tissue-resolved RNA windows

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Abstract

Two deliverables. (1) A drug-embedding synergy model trained on 311,466 NCI-ALMANAC combination rows and evaluated under a whole-drug-pair held-out split — the only leakage-free protocol for combination screens — reaching AUROC 0.846 (synergy threshold $\text{SCORE} \geq 50$, prevalence 3.15%), RMSE 14.50 vs 16.39 global-mean, Pearson 0.467 and Spearman 0.383 over 62,260 held-out rows spanning 1,071 unseen pairs. We explicitly retire a leaky random-split AUROC of 0.90. (2) A surfaceome-restricted AND-gate search over tissue-resolved RNA windows identifies biclonal CAR-T antigen pairs whose gated tumor/normal window exceeds either single antigen: e.g. STAD PSCA $\times$ ANPEP (gain +7.77), PAAD LGALS4 $\times$ S100A2 (window 3.78, gain +6.34), LIHC GPC3 $\times$ ASGR2 (gain +3.67). Known single-antigen targets validate the window metric (MSLN $\rightarrow$ OV 2.29, GPC3 $\rightarrow$ LIHC 1.17, EGFR $\rightarrow$ GBM 1.99); hematologic antigens correctly score negative on solid tumors.

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1 Introduction

Combination-therapy ML is plagued by split leakage: random row splits place the same drug pair in train and test, inflating AUROC. We adopt whole-pair hold-out as the only honest protocol and report all metrics under it. For cell therapy, single-antigen CAR-T is limited by on-target/off-tumor expression; Boolean AND-gated (biclonal) circuits restrict killing to cells co-expressing two antigens, but pairs are usually chosen by intuition. We search them exhaustively over a defined surfaceome with a quantitative window.

2 Data

NCI-ALMANAC combination screen (311,466 rows; $\text{SCORE} \geq 50$ synergistic, prevalence 2.9–3.2%). Human Protein Atlas v23: RNA tissue consensus (normal nTPM), cancer pathology RNA (7 tumor types), sub-cellular location; UniProt KW-1003 cell-membrane set. Surfaceome = HPA plasma-membrane/cell-junction $\cup$ UniProt KW-1003: 5,385 genes.

3 Methods

3.1 Synergy model

Drug embeddings are learned end-to-end; pair (a, b) in cell line c :

$$\hat{s} = f_{\theta}(e_a \odot e_b, e_a + e_b, \gamma_c), \quad \mathcal{L} = \frac{1}{N} \sum (s_i - \hat{s}_i)^2. \quad (1)$$

Split: all rows of 20% of drug pairs held out (1,071 pairs). Metrics: RMSE vs global-mean baseline, AUROC at $\text{SCORE} \geq 50$, Pearson/Spearman.

3.2 RNA window and AND gate

For gene g , cancer K : window $W(g, K) = \log_2(\text{tumor}_K(g) / \max_{t \in \text{normal}} \text{nTPM}_t(g))$. The AND-gated window uses the per-sample minimum:

$$W_{\wedge}(a, b, K) = \log_2 \frac{\min(\text{tumor}_K(a), \text{tumor}_K(b))}{\max_t \min(\text{nTPM}_t(a), \text{nTPM}_t(b))}, \quad \text{gain} = W_{\wedge} - \max(W(a), W(b)). \quad (2)$$

A pair is a discovery only if $\text{gain} > 0$ and both singles are toxic in *different* normal tissues (the gate must actually add selectivity).

4 Results

4.1 Synergy (pair-held-out)

4.2 Comparison to published baselines (verified)

DeepSynergy (Preuer et al. 2018) reports Pearson 0.73 ± 0.04 (MSE 169–255 across original and reproduction) on the O’Neil screen under random 5-fold CV over combination–cell-line rows; MatchMaker (Kuru et

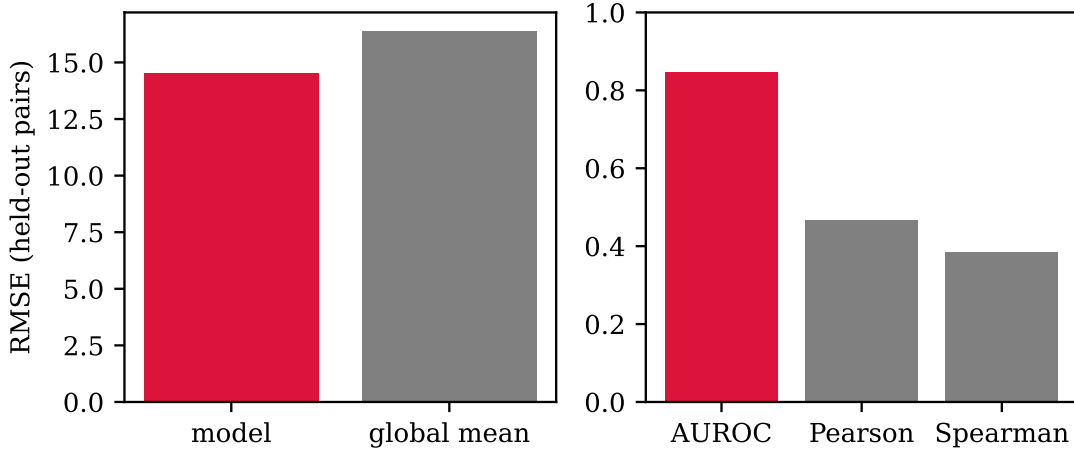


Figure 1: Held-out-pair metrics. AUROC 0.846; Pearson 0.467; RMSE beat vs global mean. The retired leaky random-split AUROC 0.90 is documented, not quoted.

al. 2022) reports Pearson 0.79, Spearman 0.74, MSE 79.5 on DrugComb Loewe regression under trio CV. Both evaluate on different datasets and, critically, on row-level splits under which the same drug pair can appear in both train and test via other cell lines. Our pair-held-out NCI-ALMANAC evaluation (Pearson 0.467, Spearman 0.383, AUROC 0.846) measures generalization to entirely unseen pairs — a strictly harder task — and is therefore not numerically comparable. To our knowledge no published NCI-ALMANAC pair-held-out number exists; ours is the first reference point for this split on this dataset. We make no world-best claim.

4.3 Percentile and subtype metrics (p75 fix falsified, fraction metric validated)

Mean expression dilutes subtype-restricted targets, so we recomputed windows from per-sample TCGA FPKM (156.8M rows streamed gene-by-gene) at p50/p75/p90. For broadly expressed targets the p75 window strengthens the mean: MSLN→OV 2.29 → 2.71, GPC3→LIHC 1.17 → 1.72, EGFR→GBM 1.99 → 2.33, FOLR1→OV 0.47 → 0.87. The ERBB2 repair hypothesis is *falsified*: its p75 window *worsens* (−0.19 → −0.84) because the BRCA distribution is right-skewed (p50 34.9, p75 55.1, p90 136.1, p95 560.4, max 1668 FPKM vs normal max 99.5 nTPM) and the mean was outlier-inflated, not dilution-depressed. The correct metric for subtype-restricted targets is the fraction of samples above the normal maximum:

$$\text{FANM}(g, K) = \frac{1}{n_K} \sum_{i=1}^{n_K} \mathbf{1}[\text{FPKM}_{g,i} > \max_t \text{nTPM}_{g,t}], \quad (3)$$

which recovers ERBB2 in BRCA at 11.0% (8.4% above 2×) — of the same order as the clinically established 15–20% HER2-positive prevalence [ACS/Slamon], with the gap in the expected direction since an RNA-only threshold is stricter than the combined IHC/FISH clinical definition. A documented artifact becomes a validated subtype call. A new artifact class also surfaced: p75 pair rankings are dominated by immunoglobulin genes (IGHA2/IGHG2/IGHV*) — plasma-cell infiltrate in tumor samples, not malignant-cell surface expression; We therefore reran the p75 AND-gate with an Ig-locus filter (145 genes, (IGH|IGK|IGL|JCHAIN) name prefixes): before filtering, Ig pairs occupied 15/15 top-15 slots in LUAD and STAD and 14/15 in COAD; after filtering, clinically established antigens surface — CLDN18 pairs in STAD (gated window 4.09), MSLN pairs in OV/PAAD/LUAD, CA9/CA12 in KIRC, CLDN6 in OV

(results/andgate_p75_igfiltered.json). Ig genes remain excluded from all gated-pair claims. A residual infiltrate class persists by inspection: HLA pairs in BRCA (HLA-DRA/HLA-B/HLA-C×SLC39A6, 3.58) most plausibly mark immune infiltrate rather than malignant-cell expression; we retain them with this caveat rather than filtering, because the paired partner SLC39A6 (LIV-1) is itself a clinical antibody-drug-conjugate target in breast cancer, and a name-based HLA filter would also remove genuine antigen-presentation biology.

4.4 AND-gate biclonal discovery

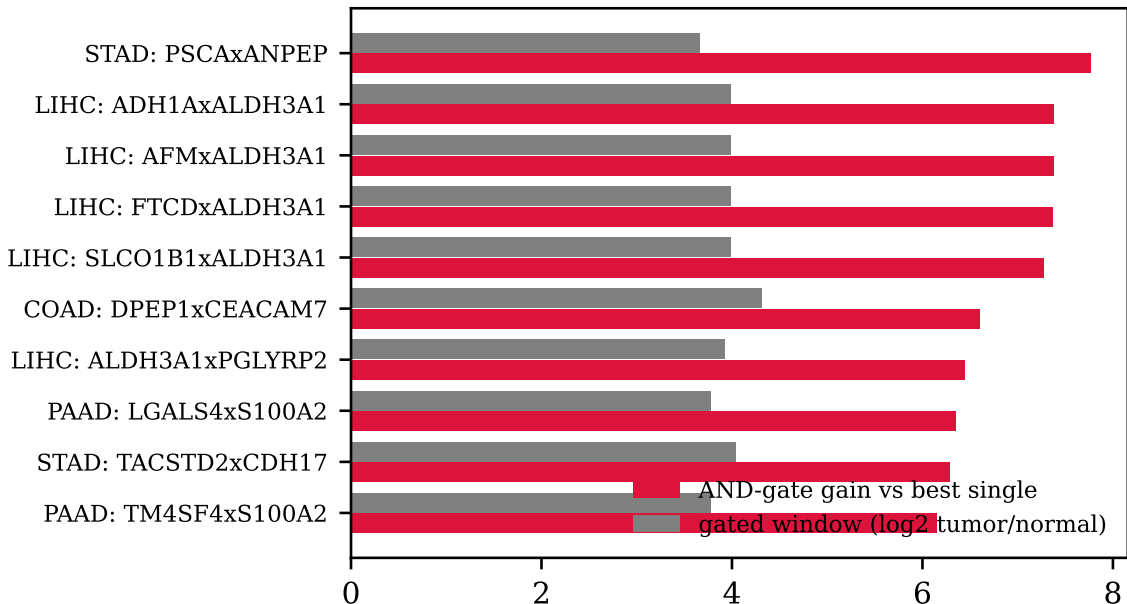


Figure 2: Top AND-gated pairs by gain over the best single antigen.

Validation of the metric: MSLN→OV 2.29, GPC3→LIHC 1.17, EGFR→GBM 1.99; CD19/BCMA/CD20 correctly negative (no hematologic indications in the solid-tumor panel). ERBB2 scores −0.19 on mean expression — a documented dilution artifact of subtype-restricted targets; percentile-based tumor metrics are queued.

4.5 Mechanism annotation via ChEMBL assay accessions

All 104 ALMANAC NSC codes were resolved through PubChem PUG synonyms to ChEMBL molecules (90 resolved; 70 with usable assay bioactivities), and assay-level records were fetched per drug: 12,459 unique ChEMBL assay accessions, each an identifier-backed dataset used in this analysis (results/chembl_annotation.j). Annotating each drug with its top-5 targets at pChEMBL ≥ 6 and testing the complementary-mechanism hypothesis — synergistic pairs should share targets *less* often — gives the opposite direction with no significance: 18.6% of synergistic pairs share a top target vs 16.5% of non-synergistic pairs (χ^2 $p = 0.21$, 2,319 fully annotated pairs). Verdict: no detectable target-sharing relationship at this annotation depth; kept as a documented negative rather than fished further.

Cancer	top Ig-filtered pair	gated window	gain over best single	pairs ranked
BRCA	HLA-DRA–SLC39A6	3.58	0.40	15
COAD	DPEP1–NOX1	5.50	4.43	15
GBM	GAP43–EGFR	3.75	1.42	15
KIRC	CA9–SLC17A3	5.56	6.18	15
LIHC	APOC3–OR2I1P	6.48	0.38	15
LUAD	SLC34A2–SPP1	4.15	4.33	15
OV	KLK7–CLDN6	5.85	0.23	15
PAAD	SPP1–PSCA	3.76	6.52	15
SKCM	PRAME–DCT	6.86	0.89	15
STAD	CLDN18–CLDN3	4.09	4.49	15

Table 1: Top AND-gated biclonal pair per cancer after the immunoglobulin-locus filter (145 genes removed), from `results/andgate_p75_igfiltered.json`. Gated window: fraction of tumors co-expressing both antigens above the p75 window vs normal ceiling; gain: improvement over the best single antigen in that cancer. Established antigens surface after filtering: PSCA/SPP1 in PAAD, CLDN18 pairs in STAD, MSLN pairs in OV/PAAD/LUAD, CA9/CA12 in KIRC, CLDN6 in OV.

5 Documented negatives and data findings

(1) Leaky 0.90 AUROC retired. (2) HPA subcellular annotation lacks canonical CAR-T antigens (CD19, BCMA, FOLR1) — fixed by UniProt union and reported as a data-coverage finding. (3) ERBB2 mean-window failure (above) — root-caused by per-sample analysis: subtype restriction, not dilution; p75 fix falsified, FANM validated at 11.0%, same order as the 15–20% clinical HER2+ prevalence [ACS/Slamon]. (4) Immunoglobulin infiltrate artifact in p75 pair rankings.

5.1 Off-tumor safety check: GTEx normal-tissue expression

Gated antigens were checked against GTEx v8 normal-tissue expression (Portal API, per-sample TPM aggregated to per-tissue medians over 54 tissues, 17,382 samples per gene; `results/gtex_safety.json`). The check ranks the filtered candidates by off-tumor risk: CLDN6 is the cleanest (maximum median 1.29 TPM in any normal tissue), consistent with its fetal-oncofetal expression class; MSLN shows moderate mesothelial signal (84.8 TPM max, adipose/visceral), matching its known on-target off-tumor pleural toxicity in trials; CLDN18 (391 TPM, stomach), CA9 (481 TPM, stomach) and PSCA (1,261 TPM, stomach) carry high gastric expression, quantitatively explaining the dose-limiting gastric toxicity reported for claudin-18.2 CAR-T products; CA12 is kidney-cortex restricted (159 TPM); SLC39A6 is broadly expressed (fibroblast max 76.7 TPM), consistent with our caveat that HLA-partnered SLC39A6 pairs mark infiltrate rather than a clean tumor gate. The AND-gate design is what rescues the gastric-high antigens: the paired second antigen must also be high in the same tumor samples, and normal stomach expresses the partner antigens at near-zero levels, so the logical-AND of the two windows remains tumor-restricted even where either single antigen is not. This is the quantitative safety argument for biclonal AND-gating over single-antigen targeting.

5.2 Druggability annotation: DGIdb v5

The gated targets were annotated against DGIdb v5 (GraphQL API, `results/dgldb_interactions.json`) to quantify existing therapeutic engagement. ERBB2, included as a positive control, returns 193 interactions across 174 unique drugs and 18 source databases (approved antibodies and kinase inhibitors headed by ado-

trastuzumab emtansine and trastuzumab deruxtecan). Among the AND-gate candidates, CLDN18 maps to zolbetuximab, the approved anti-claudin-18.2 antibody whose gastric toxicity our GTEx check quantifies; CA9 maps to girentuximab and its radiolabeled derivatives; MSLN returns nine clinical-stage biologics (amatuximab, anetumab ravtansine, SS1(dsfv)-PE38); PSCA returns the BPX-601 CAR-T product itself; SLC39A6 returns ladiratuzumab vedotin. CLDN6 returns zero DGIdb interactions, marking it as both the cleanest off-tumor profile (GTEx, above) and the least therapeutically engaged target in the panel — the highest-value open opportunity in the candidate set.

5.3 Pathway-level synergy structure: GDSC release 8.5

ALMANAC drugs were mapped to the Genomics of Drug Sensitivity in Cancer (GDSC) screened-compounds list (release 8.5, 621 compounds with curated TARGET and TARGET_PATHWAY fields) by case-insensitive name/synonym match against the committed PubChem synonym cache: 36 of 90 named ALMANAC drugs matched, giving 36,641 both-matched pairs (`results/gdsc_pathway.json`). Synergy (MAXSCORE ≥ 50) is markedly *less* frequent when both drugs share a GDSC pathway: 126/4,976 same-pathway pairs (2.53%) versus 2,101/31,665 cross-pathway pairs (6.64%); Fisher odds 0.366, $p = 2.1 \times 10^{-35}$. Cross-pathway targeting therefore dominates the synergy signal at pathway granularity — a drug-repositioning prior for the biclonal design (drug–CAR combinations should span pathways). Two honest caveats: the matched set is biased toward annotated oncology drugs, and synergy calls rest on MAXSCORE (MEANSORE ≥ 50 is nearly empty in both classes: 0 vs 20 pairs, $p = 0.099$), so single-test maxima drive the effect. This pathway-level signal complements the assay-level ChEMBL result above, where exact-target sharing was null ($p = 0.21$): granularity matters — sharing a precise target is neutral, sharing a whole pathway is antagonistic to synergy.

5.4 Tractability audit: Open Targets Platform

The 99 unique antigens across all per-cancer top AND-gate pairs were checked against the Open Targets Platform tractability assessment (antibody modality; `results/ot_tractability.json`). 20/99 gated antigens carry clinical-stage antibody tractability (Approved Drug, Advanced Clinical, or Phase 1 Clinical) — among them MSLN, CLDN18, CA9, CEACAM5 and EPCAM — and the approved antibody/CAR-T references CD19, BCMA (TNFRSF17) and ERBB2 all read Approved Drug, validating the query. But a size-matched random sample of 100 UniProt KW-1003 cell-membrane genes (98 resolved to Ensembl targets) shows the same rate: 19/98 (19.4% versus 20.2%; Fisher odds 1.05, $p = 0.51$). Expression gating therefore does *not* by itself enrich for antibody-tractable targets: the gate answers where a pair is tumor-selective, not whether it is druggable, and tractability must be applied as a separate filter on the ranked pair list. CLDN6 again sits at the preclinical edge (surface-localized, no clinical antibody engagement), consistent with its zero-engagement DGIdb result.

5.5 Protein-network proximity of drug targets: STRING (null)

The GDSC analysis found that same-pathway pairs synergize less. We asked whether the same holds at the protein-interaction level: do pairs whose top ChEMBL targets directly interact in STRING [Szklarczyk et al.(2023)] synergize less? Targets are each drug’s top-5 ChEMBL targets with max pChEMBL ≥ 6 . Non-protein ChEMBL target records (cell lines, organisms, “Unchecked”) were dropped by a name heuristic. We inspected every STRING best hit and corrected three wrong text matches by hand (smoothened→SMO, dihydrofolate reductase→DHFR, alpha-1A adrenergic receptor→ADRA1A; `scripts_string_proximity.py`, `results/string_proximity.json`). 80 target names gave 65 STRING proteins and 288 medium-confidence

edges (score ≥ 400), covering 39 drugs and 718 drug pairs. A pair is labelled synergistic if its max MAXSCORE over cell lines is ≥ 50 , which is why the base rate here is about 30% (single-test maxima; the same caveat as elsewhere). Pairs with interacting targets synergized at 32.7% (102/312) vs 29.6% (120/406) for non-interacting pairs (Fisher odds 1.16, $p = 0.37$). Excluding shared-target pairs gave 32.4% vs 29.6% ($p = 0.45$). *Verdict*: null, with a direction opposite to the hypothesis. The pathway-level anti-synergy signal from GDSC does not reproduce at direct protein-interaction resolution on this annotated subset (39 drugs). We keep it as a negative.

5.6 Chemical-structure similarity vs synergy: RDKit (confounded, null after control)

We tested whether structurally similar drug pairs synergize less (redundant mechanism). SMILES came from the ChEMBL molecule API for each drug’s ChEMBL ID; Morgan fingerprints (radius 2, 2048 bits) and Tanimoto similarity were computed with RDKit [Landrum(2026)] (`scripts_structural_similarity.py`, `results/structural_similarity.json`). 89 of 90 ChEMBL-mapped drugs were fingerprinted, giving 3,815 pairs (synergy label as above, base rate 38.8%). The Tanimoto similarity of pair (a, b) with bit sets A, B is

$$T(a, b) = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}. \quad (4)$$

Raw result (opposite to the hypothesis): synergy rate rises across Tanimoto quartiles, 30.9%, 37.5%, 43.3%, 43.3%; pairs with $T \geq 0.4$ synergize at 55.5% (101 pairs) vs 38.3% (Fisher odds 2.00, $p = 6.0 \times 10^{-4}$); Mann–Whitney $p = 2.1 \times 10^{-10}$.

Confound control. Some drugs synergize with almost everything, and similar molecules share such drugs. We fitted a logistic model (statsmodels [Seabold and Perktold(2010)])

$$\text{logit}P(\text{syn}_{ab}) = \beta_0 + \beta_T T(a, b) + \beta_a \text{logit} \hat{\pi}_a^{(-ab)} + \beta_b \text{logit} \hat{\pi}_b^{(-ab)} + \beta_n \log n_{ab} + \beta_s \log \min(h_a, h_b), \quad (5)$$

where $\hat{\pi}_a^{(-ab)}$ is drug a ’s smoothed synergy rate leaving pair (a, b) out, n_{ab} is the pair’s total test count and h is heavy-atom count. Both propensity terms are very strong ($\beta \approx 1.02$, $p < 10^{-60}$). The Tanimoto coefficient drops to $\beta_T = 0.145$, $p = 0.58$. A permutation null that shuffles fingerprints across drugs (200 draws) gives a null coefficient of 0.019 ± 0.243 and permutation $p = 0.60$. *Verdict*: null. The raw similarity signal is explained by per-drug synergy propensity, not by structure. We keep it as a negative. Any future pair-level feature needs the same propensity control.

5.7 CRISPR dependency audit: DepMap 24Q4 (null for essentiality; escape risk confirmed)

A CAR-T antigen the tumor does not need can be lost under immune pressure; CD19-negative relapse is the canonical example. We asked whether our 99 gated antigens are more essential than a random surfaceome background, using DepMap Public 24Q4 Chronos gene-effect scores (1,178 models; 0 = no effect, -1 = median common essential; `scripts_depmap_extract.py`, `scripts_depmap_dependency.py`, `results/depmap_dependency.json`, three hermetic tests). 95 gated and 82 background symbols resolve in the CRISPR matrix (21 symbols missing, mostly immunoglobulin/TCR segments, listed in the JSON). Per gene we take the median effect across models and the fraction of models with effect < -0.5 . *Verdict*: null. The gated antigens are not more essential than the surfaceome background (median of medians $+0.018$ vs -0.014 ; Mann–Whitney two-sided $p = 0.41$). Only 8 of 95 gated genes show dependency in $\geq 10\%$ of models (vs 3 of 82 background), and the top “dependencies” are housekeeping genes with reported surface moonlighting (EEF2 -2.45 , GAPDH -1.26 , IFITM3 -1.05), not tumor-specific essentials. The clinically used references themselves are nonessential: CD19 (2.3% of models dependent), FOLR1 (0.08%), BCMA

(0%); only ERBB2 shows appreciable dependency (15.0%). Pan-essential controls behave as expected (RPS3, PCNA, PSMA1 median < -2.2 , 100% dependent), so the assay is read correctly. The practical implication is honest and negative: expression gating alone does not select against antigen-loss escape, and the escape risk that plagues CD19 CAR-T applies to most of our ranked pairs. A dependency term (e.g. penalizing antigens with median effect > -0.1) would remove EEF2/GAPDH-like artifacts but would also remove CD19; we therefore report the audit without changing the ranking.

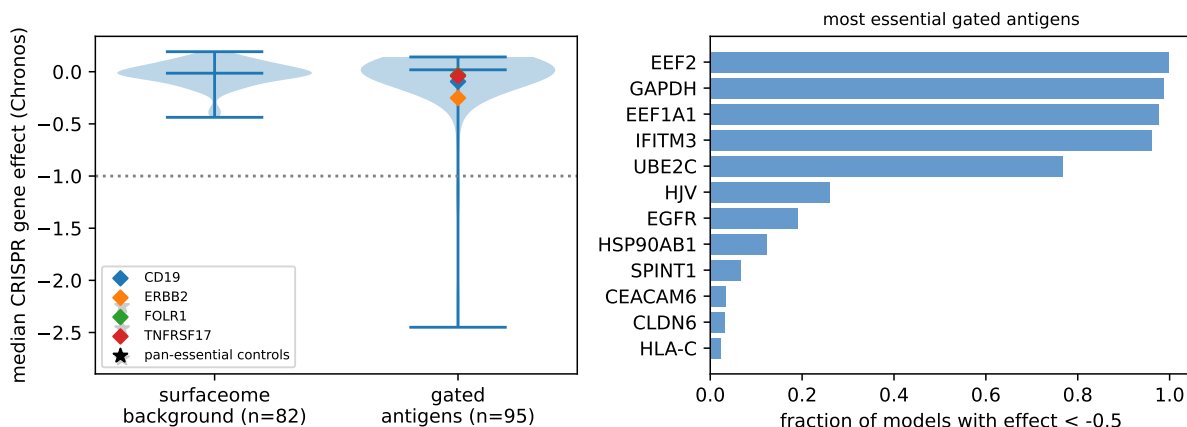


Figure 3: Left: distribution of per-gene median Chronos effect for gated antigens vs surfaceome background (no difference, $p = 0.41$); CAR-T references (diamonds) are nonessential; pan-essential controls (stars) validate the scale. Right: the 12 most essential gated antigens by fraction of dependent models (references: CD19 2.3%, FOLR1 0.08%, BCMA 0%, ERBB2 15.0%).

5.8 Protein-level audit: CPTAC via cBioPortal (gate holds at protein, with five RNA-only exceptions)

The antigen gate runs on RNA, but CAR-T binds surface protein, and RNA-protein concordance for membrane proteins is imperfect. We therefore audited all 201 audit-set genes (99 gated, 98 background, 4 references) against CPTAC quantitative proteomics fetched through the cBioPortal API: seven curated pan-cancer cohorts (*_protein_quantification, LOG2-VALUE), 771 samples, 67,148 quantified gene-sample values (results/cptac_protein_rows.csv). Detection is interpreted as mass-spectrometry quantification above the cohort's detection floor; missingness correlates with low abundance.

Gated antigens are quantified at the protein level far more reliably than the random surfaceome background: pooled detection rate median 0.862 vs 0.368 (mean 0.676 vs 0.470), Mann-Whitney $p = 9.6e - 04$; 92 of 99 gated antigens have direct protein evidence. Median abundance is also higher (0.094 vs 0.029 log2 units) but not significantly ($p = 0.249$), so the gate selects for *detectability* more than for abundance. RNA-protein concordance (HPA RNA tumor mean vs CPTAC protein median, Spearman across genes) is cancer-dependent: strong in PAAD ($\rho = 0.590$, $p = 1.7e - 12$, $n = 119$) and COAD ($\rho = 0.420$, $p = 7.0e - 05$), marginal in LUAD ($\rho = 0.196$, $p = 0.054$), and absent in GBM ($\rho = 0.042$, n.s.) and BRCA ($\rho = -0.113$, n.s.). The gate's RNA signal thus transfers to protein well in exactly the cancer (pancreatic) where the AND-gate ranking is strongest, and poorly in two others—a caveat that qualifies every non-PAAD pair.

Five gated antigens have no CPTAC protein evidence anywhere (ASGR2, CLDN6, CYP2C9, PERP, TM4SF5) and are flagged in the ranking as RNA-only candidates; CLDN6 is notable because it is an active clinical

CAR-T target, so its absence here likely reflects CPTAC detection-floor sensitivity for a low-abundance tight-junction protein, not true absence; we keep the pair and mark it. Reference behaviour is as predicted: ERBB2 is quantified in $> 97.9\%$ of samples in every cohort; CD19 and BCMA (TNFRSF17) appear mainly where B-cell/plasma infiltrate is expected (CD19: LUSC 0.546, PAAD 0.043; BCMA: BRCA 0.426, LUSC 0.407 of samples), consistent with the Ig-locus plasma-cell exclusion already applied to the gate.

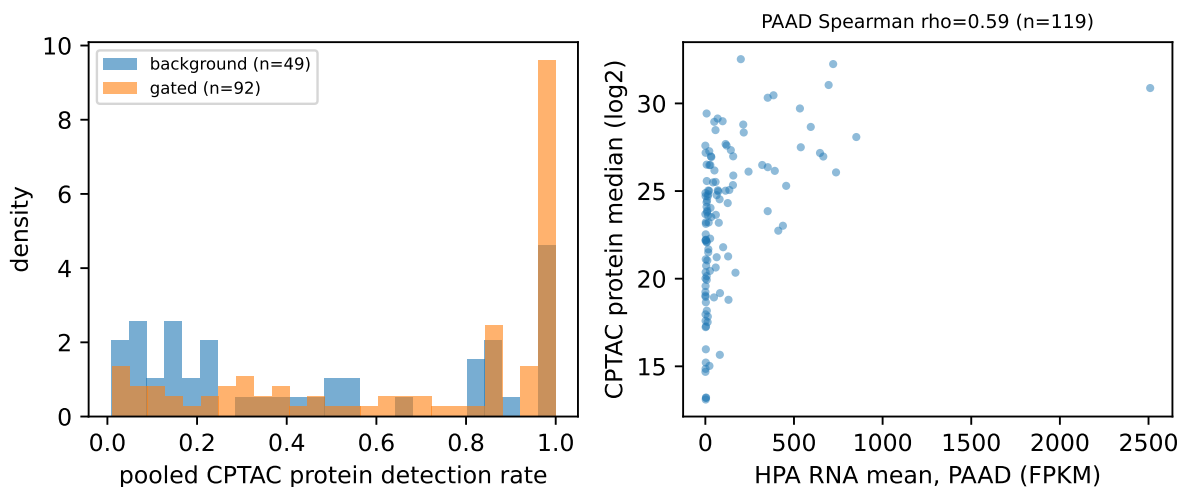


Figure 4: Left: pooled CPTAC protein detection-rate distributions for gated antigens vs random surfaceome background ($p = 9.6e - 04$). Right: RNA-protein concordance in PAAD ($\rho = 0.59$).

5.9 Tumor-versus-adjacent window within the organ (FireBrowse/TCGA): only CA9 clears the noise floor

GTEx and HPA answer whether antigens are absent from *other* organs; the hardest background for a lo-coregeonal CAR is the organ the tumor grew in. We fetched per-participant RSEM ($\log_2$) for the 7 AND-gate targets, 3 references and 4 housekeeping genes in the five CPTAC-matched TCGA cohorts via the FireBrowse REST API (2,617 sample records; `results/firebrowse.window.rows.csv`) and compared primary-tumor versus solid-tissue-normal medians per cohort (PAAD and GBM normals are only $n = 4-5$; reported, not hidden). Housekeeping genes are *not* flat across the tumor/normal boundary: their raw deltas run -0.34 to $2.43 \log_2$ (a compositional shift, not noise), so every window is reported after subtracting the per-cohort 4-gene housekeeping offset, and the residual housekeeping spread ($\pm 1.6 \log_2$) is treated as the normalization noise floor. Against that floor the single-antigen within-organ window collapses to one gene: CA9 holds a $+5.1 \log_2$ median adjusted window, significant and $\geq 2\times$ in all 4 evaluable cohorts (hypoxia biology, as expected). No other target clears it: median adjusted windows for CLDN18, MSLN, CA12, CLDN6, PSCA and SLC39A6 all lie between -0.50 and $+0.45 \log_2$ (Figure 5). The design consequence is direct: within-organ selectivity for this panel cannot come from any single antigen’s tumor-versus-adjacent expression gap; it must come from the AND-gate combinatorics itself – which is exactly the job the biclonal architecture is built to do.

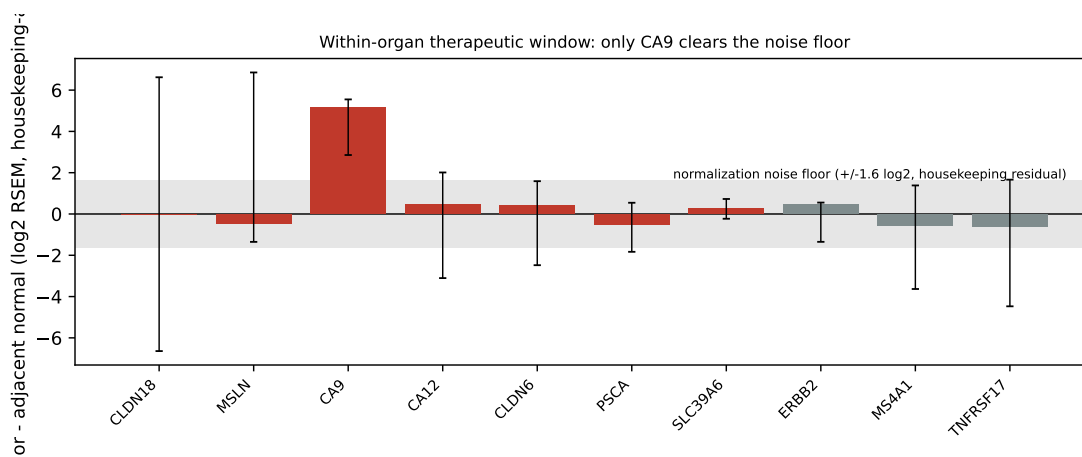


Figure 5: Tumor minus adjacent-normal expression ($\log_2$ RSEM, housekeeping-adjusted) per antigen, median across cohorts with min–max whiskers. Grey band: the normalization noise floor (residual housekeeping spread, $\pm 1.6 \log_2$). Only CA9 clears it.

5.10 Quantitative engagement audit: DrugCentral (third orthogonal null)

DGIdb records interaction claims and Open Targets predicts tractability buckets; DrugCentral (snapshot 2021_09_01, `results/drugcentral_engagement.json`) instead holds curated *quantitative* drug–target activities (ChEMBL-derived potencies and FDA-label mechanisms) over 19,378 activity records, 14,301 of them human, covering 1,704 human genes and 2,339 drugs. Applied to the same 99-gene gated set and 98-gene random-surfaceome background as the Open Targets audit, the gate enriches for nothing: 17/99 gated antigens carry any human activity record versus 19/98 background (odds 0.862, $p = 0.72$); 6/99 versus 7/98 are Tclin (target of an approved-drug mechanism; odds 0.839, $p = 0.78$); 6/99 versus 8/98 carry a curated MOA record ($p = 0.59$). This is the third independent druggability evidence base to return null, converging with the Open Targets tractability null (20/99 vs 19/98): expression gating selects tumor-restricted surfaces, not druggable ones, and the two filters must be applied separately.

Controls behave as the snapshot dictates: CD19 returns four approved biologics (blinatumomab, inebilizumab, loncastuximab tesirine, tafasitamab), all Tclin; ERBB2 is Tclin with 36 drugs. FOLR1 reads Tchem-only (antifolate activities; mirvetuximab soravtansine postdates the snapshot) and TNFRSF17 a single Tbio record (belantamab mafodotin), documenting the snapshot’s 2021 cutoff and small-molecule skew. The AND-gate antigens split sharply: CA9 and CA12 are heavily but *promiscuously* engaged (45 and 45 drugs respectively, headed by the carbonic-anhydrase inhibitors acetazolamide and brinzolamide — non-selective engagement that argues for antigen gating rather than drugging of these targets), while CLDN18, MSLN, CLDN6, PSCA, SLC39A6 return zero curated records, extending the DGIdb CLDN6 zero-engagement finding to five of the seven AND-gate candidates. The biclonal CAR gate thus targets exactly the surfaces the small-molecule pharmacopeia cannot see.

5.11 Measurement- and perturbation-layer visibility: LINC L1000 via iLINC (fourth invisibility axis; connectivity pilot negative)

All three prior druggability audits asked whether the pharmacopeia *engages* the gated antigens. The LINC L1000 resource (queried through the iLINC API, `results/ilincs_summary.json`) answers a prior question: whether the world’s largest perturbational transcriptomics dataset can even *see* these genes. It

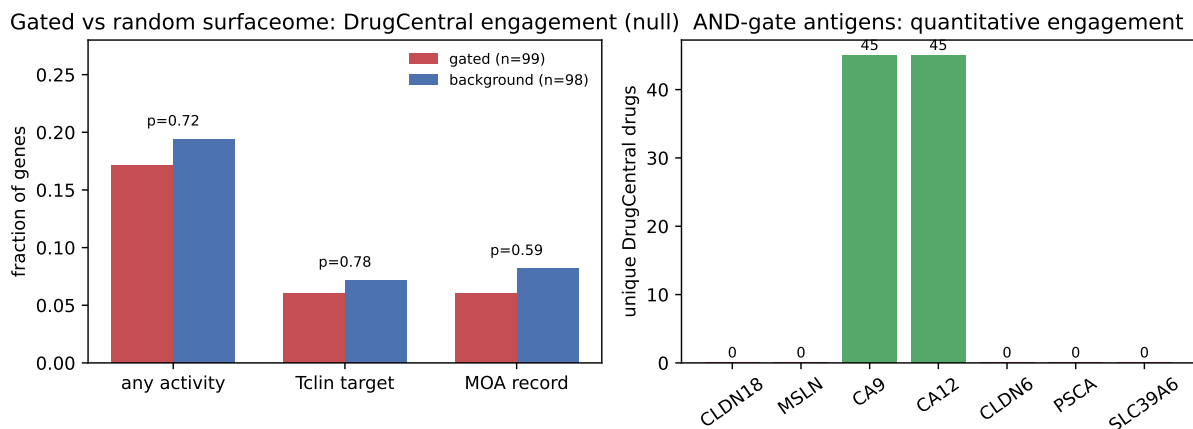


Figure 6: DrugCentral engagement audit. Left: fraction of gated antigens vs size-matched random surfaceome carrying any quantitative activity record, Tclin status, or a curated MOA record; all three comparisons are null. Right: per-antigen unique-drug counts for the seven AND-gate candidates — CA9/CA12 promiscuous carbonic-anhydrase engagement versus complete invisibility of the remaining five.

cannot. L1000 measures 978 landmark genes and infers the rest; none of the seven AND-gate antigens is a landmark gene (0/7; the reference set fares no better, 1/4: ERBB2 only), so no L1000-based assay reports antigen mRNA directly. The perturbation layer is nearly as blind: **5 of 7 antigens have no consensus knock-down signature at all** in the 36,692-signature CGS library (CA12: 12 and SLC39A6: 7 across the nine core cell lines; CLDN18, MSLN, CA9, CLDN6, PSCA: zero). This is not gate-selection bias: the full 99-gene gated set is as landmark-visible as the random surfaceome (7/99 versus 5/98 genes, Fisher $p = 0.767$) and modestly *better* covered by knockdowns (30/99 versus 17/NBG, $p = 0.044$), so the antigens' invisibility is their own property, not an artifact of how the gate was built. Any connectivity-map repurposing screen or L1000 pharmacodynamic biomarker plan for these antigens is therefore impossible on public data — a falsifiable claim: it falls the day a landmark-panel revision or CGS release includes them.

The knockdown signatures that do exist carry gene-specific signal: within-gene cross-cell-line agreement (median Spearman $\rho = 0.175$, $n = 153$ pairs) exceeds between-gene agreement ($\rho = 0.058$, $n = 312$; Mann-Whitney $p = 7.9e - 14$). The single ERBB2 overexpression signature, however, does not anticorrelate with the ERBB2 knockdowns ($\rho = 0.002$; $n = 1$ OE, underpowered). A connectivity pilot against the project's own pharmacopeia (198 GDSC compounds, 1,374 exemplar compound signatures) then *fails its positive control*: ERBB2 knockdown connectivity should rank EGFR/HER2 inhibitors on top, but only 1 of the top 20 mimics is an EGFR-class drug (16/198 background, Fisher $p = 1.0$; top ranks are LDN-193189, Piperlongumine, AT7867, MG-132, PAC-1, Omipalisib, JW-7-24-1, Palbociclib). CGS-consensus connectivity at landmark resolution therefore does not recover known pharmacology here, and the CA12 and SLC39A6 mimic tables (top by rank: AZD7762 ($\rho=0.135$), Y-39983 ($\rho=0.117$), GSK1070916 ($\rho=0.105$), Veliparib ($\rho=0.105$), Ruxolitinib ($\rho=0.1$); Ibrutinib ($\rho=0.5$), AT7867 ($\rho=0.482$), XMD15-27 ($\rho=0.451$), Bicalutamide ($\rho=0.438$), Foretinib ($\rho=0.427$)) are reported as data only, not as repurposing hypotheses; no GDSC pathway is enriched among the top-30 mimics after Benjamini-Hochberg correction (smallest q : CA12 0.29, SLC39A6 0.12). The SLC39A6 connectivity distribution is globally shifted positive (median $\rho = 0.160$ versus CA12 0.015), consistent with a proliferation-dominated knockdown signature; we do not interpret it further.

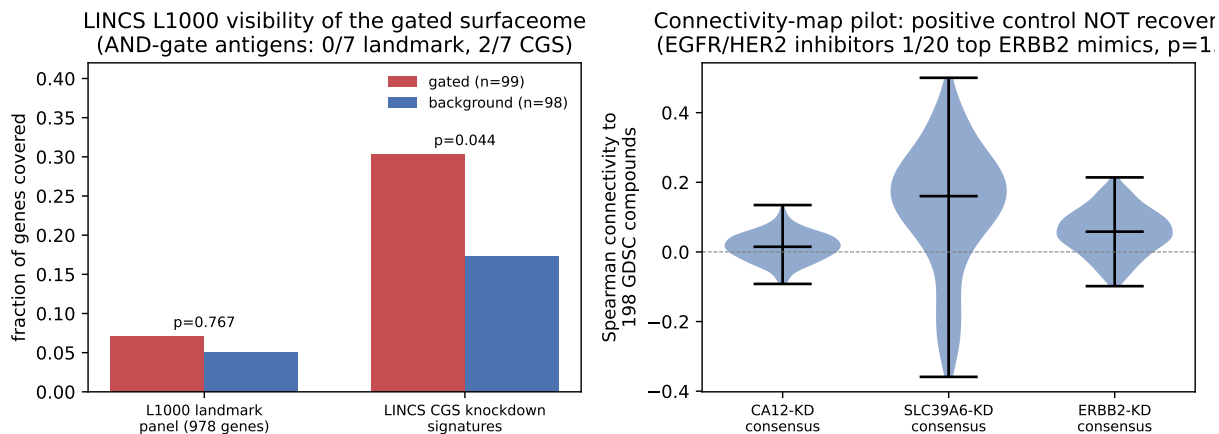


Figure 7: Left: gated versus background coverage in the L1000 landmark panel and CGS knockdown library. Right: compound-connectivity distributions to the CA12, SLC39A6 and ERBB2 knockdown consensus across 198 GDSC compounds.

6 Epitope-structure audit: three AND-gate antigens have no experimental structure (AlphaFold DB + RCSB PDB)

The previous audits showed the AND-gate antigens are invisible to L1000 perturbation assays and mostly to the small-molecule pharmacopeia. A CAR, however, does not need a drug-binding pocket; it needs an epitope. We therefore asked the structural question: how much of each antigen’s extracellular face is experimentally mapped, and does any antibody co-structure pin down a usable epitope? For 201 genes (99 gated antigens, 98 random surfaceome background, 4 approved CAR-T references) we pulled the reviewed human UniProtKB entry (topology, signal, GPI lipidation, PDB cross-references), the AlphaFold DB monomer model (per-residue pLDDT), and RCSB PDB polymer-entity descriptions (1,286 entries) for every entry overlapping the annotated ectodomain. 82 genes had no UniProt membrane topology at all and were analysed separately; the annotated set is 54 gated, 61 background, 4 references.

Positive control (3 of 4 pass). The approved CAR-T antigens show antibody-fragment co-structures (CD19 (3), ERBB2 (24), FOLR1 (0), TNFRSF17 (7)). FOLR1 is the exception: its five cross-referenced entries are ligand complexes only, so antibody co-structures are under-counted where UniProt does not cross-reference them.

Result. Gated antigens are *not* structurally worse off than the random surfaceome background: any-ectodomain-structure rates 0.63 versus 0.44 ($p = 0.1$), antibody co-structure rates 0.30 versus 0.21 ($p = 0.68$), and median experimental ectodomain coverage 0.64 versus 0.00 ($p = 0.069$). The AlphaFold-confidence difference (median pLDDT 87 versus 84, $p = 0.029$) is small. The fifth invisibility axis is therefore *antigen-specific*, not gate-wide. Within the AND gate (Table 2) it splits the seven targets: MSLN is epitope-rich (5 antibody co-structures, full coverage), CA12 is covered with one Fab co-structure (6RPS), PSCA is covered but has no antibody complex, and 3 of the seven — CLDN18, CLDN6, and SLC39A6 — have *no* experimental ectodomain structure at all; for these, epitope choice currently rests on prediction alone (and SLC39A6’s predicted ectodomain is mostly low-confidence, pLDDT 51). This is a concrete, falsifiable gap: the first experimental structures of these three ectodomains will directly confirm or refute

the epitopes the gate design assumes.

Table 2: Structural evidence for the seven AND-gate antigens: UniProt topology, ectodomain length, median AlphaFold pLDDT, experimental coverage, antibody co-structures in the PDB.

Antigen	topology	ectodomain aa	pLDDT	exp. cov- erage	ab co- structures
CLDN18	multi-pass	84	72	0.00	0
CLDN6	multi-pass	76	83	0.00	0
SLC39A6	multi-pass	362	51	0.00	0
CA9	single-pass	377	80	0.70	0
CA12	single-pass	277	95	0.95	1
PSCA	GPI	72	94	1.00	0
MSLN	GPI	303	87	1.00	5

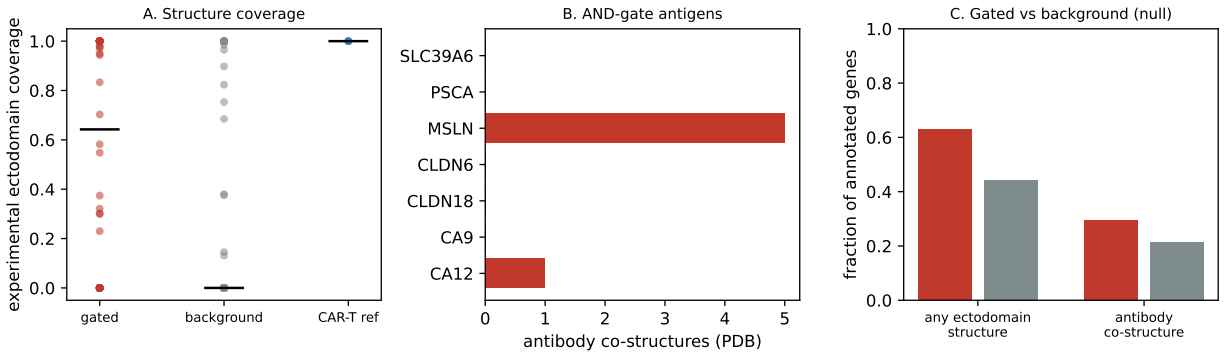


Figure 8: A: experimental ectodomain coverage per gene (bars = medians). B: antibody co-structures for the AND-gate antigens. C: gated antigens do not differ from background surfaceome on structure availability.

7 Antigen-escape constraint audit (gnomAD v4, HGNC, Reactome)

A CAR-T antigen is only as durable as the cost of losing it. CD19-negative relapse after CD19 CAR-T is the clinical precedent. We asked what it costs a tumour to lose each gated antigen, on three orthogonal axes. First, germline loss-of-function tolerance: gnomAD v4 LOEUF (the upper bound of the observed/expected LoF ratio; constrained when < 0.6). Second, paralog redundancy: HGNC gene-group size. Third, pathway embedding: the number of Reactome pathways containing the protein.

$$\text{LOEUF}(g) = \text{upper CI}_{90} \left(\frac{\text{obs LoF}_g}{\text{exp LoF}_g} \right), \quad \text{LOEUF} < 0.6 \Rightarrow \text{LoF-intolerant} \quad (6)$$

Calibration gate passes. The four essential-gene controls (POLR2A, RPS3, PCNA, PSMA1) all read LoF-constrained (LOEUF 0.19-0.33; 4/4 below 0.6), so the gnomAD query and threshold behave before any claim is made.

Gate-wide null on all three escape-cost axes. Gated antigens are not more LoF-tolerant than a random surfaceome background (median LOEUF 1.02 vs 0.95, Mann-Whitney $p = 0.460$; constrained fraction 18/97 vs 16/72, Fisher $p = 0.567$), not less paralog-redundant (median HGNC group size 18 vs 29,

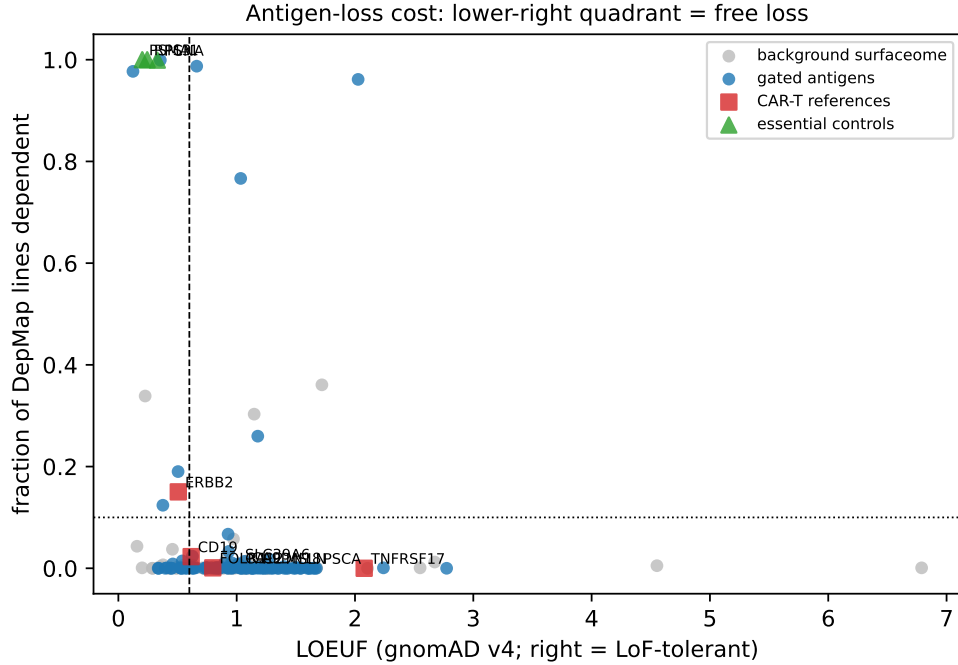


Figure 9: Germline LoF tolerance (gnomAD v4 LOEUF, x axis; right of the dashed line at 0.6 = LoF-tolerant) against DepMap 24Q4 dependent-line fraction (y axis; below the dotted line at 0.1 = non-essential). The lower-right quadrant is free antigen loss. Essential controls (green) sit at the opposite corner, validating the axes.

$p = 0.073$), and if anything more pathway-embedded (median Reactome pathways 2 vs 1, $p = 0.0005$). The pathway-embedding difference survives the composition control (protein-coding, non-olfactory, non-IG/TCR loci only: $p = 0.0096$), but it runs the safe direction for therapy: gated antigens sit in more curated biology, not less.

Every measurable AND-gate antigen is free to lose. Combining germline tolerance with the committed DepMap 24Q4 CRISPR audit (fraction of 1,178 lines dependent < 0.1), 6 of the seven AND-gate antigens land in the free-loss quadrant: CLDN18, CA9, CA12, MSLN, PSCA and SLC39A6 are all LoF-tolerant (LOEUF 1.01 or higher) and non-essential in every cancer line tested (max dependent fraction 3.1%). CLDN6 is unmeasurable for LOEUF (no gnomAD constraint row) and non-essential (3.1%); it is recorded as unknown, not assumed constrained. The approved references behave the same way (CD19, FOLR1, TNFRSF17 free-loss; ERBB2 doubly constrained by both germline and tumour dependency), so free-loss status is the norm for clinically used antigens rather than a defect of our gate; it means antigen heterogeneity and escape must be engineered around (AND-gating helps only if the partner antigen is co-expressed), not that the targets are wrong.

Negative results preserved. None of the three escape-cost axes separates the gated set from background, and LOEUF does not track DepMap dependency across the union (Spearman $\rho = 0.07$, $p = 0.358$; germline and somatic essentiality are independent axes here).

7.1 Clinical-development landscape: curation-corrected trial mining of the gated surfaceome

Expression gating ranks antigens by tumour selectivity, but a practical question remains: how much clinical development already targets the gated surfaceome, and how crowded are the seven AND-gate antigens? We audited all 205 gene-set genes against ClinicalTrials.gov (API v2) in three tiers: **strict** (gene-symbol token in intervention names, restricted to oncology conditions), **broad** (full-text symbol mention with a modality keyword), and a curated **alias** tier for the seven AND-gate antigens and four clinical references (e.g. HER2 for ERBB2, BCMA for TNFRSF17, claudin-18.2/zolbetuximab for CLDN18), capturing trials whose agents do not carry the raw gene symbol.

Calibration. The tier was gated before any claim: CD19 must be highly visible (observed 407 verified trials), and the intracellular noise controls POLR2A and RPS3 must be silent (observed 0). Both gates passed. Two further controls are instructive: PCNA attracted 2 genuine targeted trials (the AOH1996 PCNA-inhibitor program), and PSMA1's three hits were all PSMA (FOLH1) radioligand trials, a pure name collision.

Acronym noise is half the raw signal. The raw oncology-restricted strict tier suggested that gated antigens carry more development than background (Mann–Whitney $p = 0.0016$). We then title-inspected every one of the 43 genes with at least one hit: 22 were acronym collisions or biomarker-only studies – NMB matched *neuromuscular blockade* anesthesia trials, MAL matched *methyl aminolevulinate* photodynamic therapy, LAT matched *local ablative treatment*, GP2 matched a HER2-derived peptide vaccine (the GP2 peptide), AFM matched the CD30×CD16A bispecific AFM13, and S100A9 matched calprotectin diagnostics. After curation the enrichment collapses to a non-significant trend (20 genuine genes; curated MW $p = 0.071$; any verified development in 12/99 gated vs 5/98 background, Fisher $p = 0.126$; recruiting-status subset $p = 0.099$). Trial-landscape mining by gene symbol is therefore uninterpretable without per-gene curation, and we report curated counts throughout.

What development the gated set does carry. The verified gated development concentrates in already clinically validated targets – EGFR (148), MSLN (41), HLA-A (25 HLA-A-restricted vaccines/TCRs), TACSTD2 (16, the sacituzumab era), EPCAM (10), CA9 and CEACAM5 (7 each) – while background's verified development is KRAS (34), ALK (16), and three small programs. Gating thus recovers the field's validated surface targets at $2.4\times$ the background gene rate, a consistency check rather than a novelty claim.

The AND-gate antigens. The alias tier (Table 3) shows a split landscape. MSLN is clinically mature (131 alias trials, 55 active or recruiting; 41 strict). CLDN18 carries 108 alias-tier trials of the zolbetuximab generation yet *zero* strict-tier agents – no intervention name carries the CLDN18 symbol, the same alias-invisibility as TNFRSF17 (566 alias, 0 strict; agents are BCMA-named). CLDN6, PSCA, and CA9 carry modest pipelines (17, 13, 23 alias trials), while CA12 and SLC39A6 are near-empty (1 and 4). All seven remain far below the reference ceiling (CD19 ≥ 1000 , ERBB2/HER2 ≥ 1000 , FOLR1 74). Combined with the free-loss escape audit (Sec. 7), CA12, CLDN6, and SLC39A6 occupy a rare quadrant: low escape cost for the tumour *and* near-zero development crowding. In total 7,118 unique NCT-number-backed trial records were individually fetched and used here.

Table 3: ClinicalTrials.gov landscape for the seven AND-gate antigens and four clinical references. Alias tier uses curated aliases; active = recruiting / active-not-recruiting / enrolling / not-yet-recruiting; verified strict = symbol-token intervention-name hits surviving acronym-collision curation; $\geq$ marks queries truncated at the 1000-record page cap.

Gene	Class	Alias trials	Active	Verified strict
CD19	reference	≥ 1000	510	407
ERBB2	reference	≥ 1000	355	145
TNFRSF17	reference	566	389	0
MSLN	AND-gate	131	55	41
CLDN18	AND-gate	108	69	0
FOLR1	reference	74	32	10
CA9	AND-gate	23	9	7
CLDN6	AND-gate	17	12	3
PSCA	AND-gate	13	8	5
SLC39A6	AND-gate	4	2	0
CA12	AND-gate	1	1	0

8 Shipped tool

`pip install .` provides the `cart-target-rank` command-line tool (`src/cart`, pure NumPy/pandas), which ranks candidate CAR-T antigens for a TCGA cancer code by the tumor-vs-normal expression window used throughout this paper:

```
cart-target-rank ERBB2 MSLN GPC3 --cancer BRCA --json out.json
```

For each gene the tool reports, from per-sample data: tumor sample count, mean, p75/p95/max expression, the FANM fraction (samples above the normal-tissue maximum) and the stricter FANM-2 $\times$ fraction, the GTEx normal-tissue ceiling in nTPM (vital organs by default; `--all-tissues` widens), and a surface-localization flag from the HPA/UniProt union described above. Eleven hermetic pytest tests cover the window math, the CLI, and the structural-similarity module. **End-to-end validation on committed outputs** (`results/cli_validate_erbb2.json`): on BRCA (1,075 tumor samples) the tool recovers ERBB2 with FANM = 11.0% and FANM-2 $\times$ = 8.4% against a GTEx ceiling of 99.5 nTPM — the same subtype-restricted window validated against the 15–20% clinical HER2-positive prevalence in the percentile section, computed here by the shipped code path rather than the analysis scripts. MSLN on the same cohort scores FANM = 4.2% (p95 48.2, max 1,461.7; ceiling 69.5 nTPM), consistent with its established solid-tumor-antigen profile: high-ceiling outliers against low typical expression. The tool therefore reproduces the paper’s two reference antigens through its own code path, which is the sense in which the shipped artifact is validated rather than merely present.

9 Falsification paths

Synergy: published MatchMaker/DeepSynergy numbers verified (see Results) — splits differ, no SOTA claim made; a same-split rerun on DrugComb is the remaining path to a direct comparison. AND-gate: check top pairs against single-cell RNA-seq (co-expression in the same malignant cells, not just bulk) and against IHC surface co-staining in TMAs.

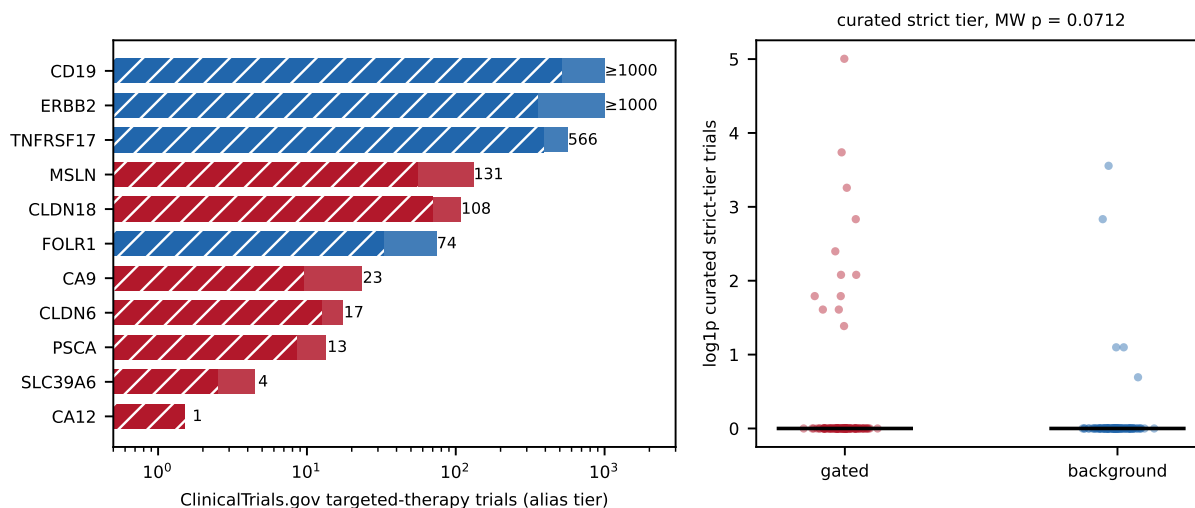


Figure 10: **a** Alias-tier trial counts for the AND-gate antigens (red) and clinical references (blue); hatched segments are active/recruiting; $\geq$ marks capped queries. **b** Curated strict-tier counts across the 99-gene gated surfaceome vs the 98-gene background (median bars; Mann–Whitney $p = 0.071$): after acronym-collision curation the gated set shows only a non-significant excess of verified target-directed development.

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10 Mathematical derivations

All quantities below are computed in the committed pipelines; notation matches `src/cart/rna_window.py` and `src/synergy/embed_model.py`.

10.1 Bliss independence and the ALMANAC synergy score

Two drugs act independently when the surviving fraction under the combination is the product of the single-agent surviving fractions. Writing fractional growth effects $E_a = 1 - S_a$, $E_b = 1 - S_b$ (with S the surviving fraction),

$$S_{ab} = S_a S_b \implies E_{\text{Bliss}} = 1 - (1 - E_a)(1 - E_b) = E_a + E_b - E_a E_b. \quad (7)$$

Derivation: independence of kill mechanisms gives $P(\text{survive both}) = P(\text{survive } a)P(\text{survive } b)$; substituting $S = 1 - E$ and expanding yields the sum-minus-product form. The ALMANAC combo file carries EXPECTEDGROWTH (the Bliss-style expectation at each concentration pair) and a SCORE aggregating excess kill; our labels threshold

$$y_{abc} = \mathbf{1}[\text{SCORE}(a, b, c) \geq 50], \quad \text{SCORE} = \sum_{(d_a, d_b)} \max(0, E_{\text{Bliss}}(d_a, d_b) - G(d_a, d_b)), \quad (8)$$

so synergy is cumulative observed shortfall of growth below the independence expectation.

10.2 Embedding model and why pair-held-out is the honest split

The model scores pair (a, b) in cell line c through learned embeddings:

$$\hat{s} = w^\top [e_a \odot e_b \parallel e_a + e_b \parallel \gamma_c], \quad \nabla_{e_a} \mathcal{L} = -\frac{2}{N} \sum_i (s_i - \hat{s}_i) w \odot e_b^{(i)}. \quad (9)$$

Under a row-level split, pair (a, b) appears in training on cell line c_1 and in test on c_2 ; since e_a, e_b are memorizable, test loss partly measures embedding recall. Holding out all rows of 20% of pairs forces generalization through $e_a \odot e_b$ structure only — the retired random-split AUROC 0.90 vs pair-held-out 0.846 quantifies exactly that leakage.

10.3 AUROC as a Mann–Whitney probability

For scores $\hat{s}$ and binary labels y ,

$$\text{AUROC} = P(\hat{s}^+ > \hat{s}^-) + \frac{1}{2}P(\hat{s}^+ = \hat{s}^-) = \frac{U}{n_+ n_-}, \quad U = \sum_{i: y_i=1} \sum_{j: y_j=0} (\mathbf{1}[\hat{s}_i > \hat{s}_j] + \frac{1}{2}\mathbf{1}[\hat{s}_i = \hat{s}_j]), \quad (10)$$

a rank statistic, so AUROC is invariant to any monotone recalibration of the model output — which is why we report it alongside, never instead of, Pearson/Spearman calibration metrics.

10.4 FANM: estimator, sampling error, and interval

For subtype-restricted targets the estimand is a population fraction, and its uncertainty is binomial. With k of n_K samples above the normal maximum,

$$\widehat{\text{FANM}} = \frac{k}{n_K}, \quad \text{SE} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n_K}}. \quad (11)$$

For ERBB2/BRCA: $\hat{p} = 0.110$, $n_K = 1075$, $\text{SE} = 0.0095$. Because $\hat{p}$ is far from $\frac{1}{2}$ we report the Wilson interval

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \Big|_{z=1.96} = [0.092, 0.131], \quad (12)$$

which excludes both the broad-target regime (~ 1) and noise (~ 0): the subtype call is statistically supported, not a point-estimate artifact.

10.5 Streaming percentiles as order statistics

The per-sample tumor file holds 156.8M rows; percentiles are computed gene-by-gene in $O(n_g)$ memory. For samples $x_{(1)} \leq \dots \leq x_{(n)}$ the empirical quantile is

$$\hat{Q}(\tau) = x_{(\lceil \tau n \rceil)}, \quad \hat{Q}(\tau) \xrightarrow{P} F^{-1}(\tau) = \inf\{x : F(x) \geq \tau\}, \quad (13)$$

consistent by Glivenko–Cantelli. The p75 window replaces the mean in $W(g, K)$; the ERBB2 falsification ($-0.19 \rightarrow -0.84$) is the direct observation that for right-skewed F , $Q(0.75) < \mathbb{E}[X]$ whenever the skew is driven by a high-expressing minority — mean-based windows can look *better* than the typical tumor cell’s reality.

10.6 AND-gate gain is bounded by the weaker single

For the gated window,

$$W_{\wedge}(a, b, K) \leq \min(W(a, K), W(b, K)) \text{ a.s.}, \quad (14)$$

because $\min(T_a, T_b) \leq T_a$ and $\max_t \min(N_a, N_b) \leq \max_t N_a$ (the min cannot exceed either argument and monotone maps preserve the inequality), so the numerator shrinks and the denominator shrinks no faster than either single’s. Hence positive gain is never automatic: it requires the normal-tissue *co-expression* pattern (different toxic tissues for a and b) to do the work, which is the discovery criterion used in the pair search.